MARKING SCHEME

Chapter 1: Chemical Reactions and Equations — Class 10 Science

Model Answers · Key Points · Chemical Equations

Instructions: Award marks if the answer contains the key points listed. Chemical equations must be balanced for full marks. Partial credit may be awarded for partially balanced equations.

EASY **Q1. Define a chemical reaction.** **[1 Mark]**

Model Answer:

A chemical reaction is a process in which one or more substances (reactants) are converted into one or more different substances (products) with a change in chemical composition. Indicators include: change in colour, evolution of gas, change in temperature, formation of precipitate.

Award marks if answer contains:

- ◆ **Process of conversion of reactants to products**
- ◆ **Change in chemical composition**
- ◆ **Any two indicators of chemical reaction**

EASY **Q2. What happens to the mass of reactants and products in a chemical reaction?** **[1 Mark]**

Model Answer:

According to the Law of Conservation of Mass: Mass of reactants = Mass of products. The total mass remains unchanged during a chemical reaction.

Award marks if answer contains:

- ◆ **Law of Conservation of Mass**
- ◆ **Total mass is conserved**
- ◆ **Mass of reactants = Mass of products**

EASY **Q3. Name the type of reaction that takes place when magnesium ribbon burns in air.** **[1 Mark]**

Model Answer:

Combination reaction (also called oxidation reaction). Magnesium combines with oxygen in air to form magnesium oxide.

Award marks if answer contains:

- ◆ **Combination reaction**
- ◆ **Oxidation reaction (accept both)**

Chemical Equation(s):

$$2Mg + O_2 \rightarrow 2MgO$$

EASY **Q4. What is a balanced chemical equation? Why should a chemical equation be balanced?** [1 Mark]

Model Answer:

A balanced chemical equation has equal numbers of atoms of each element on both sides of the equation. It must be balanced to satisfy the Law of Conservation of Mass.

Award marks if answer contains:

- ◆ Equal atoms on both sides
- ◆ Satisfies Law of Conservation of Mass

EASY **Q5. What are the physical states of reactants and products represented by the symbols (s), (l), (g) and (aq) in a chemical equation?** [1 Mark]

Model Answer:

(s) = solid, (l) = liquid, (g) = gas, (aq) = aqueous solution (dissolved in water).

Award marks if answer contains:

- ◆ (s) solid
- ◆ (l) liquid
- ◆ (g) gas
- ◆ (aq) aqueous solution

EASY **Q6. What is a combination reaction? Give one example.** [1 Mark]

Model Answer:

A reaction in which two or more substances combine to form a single new substance is called a combination reaction.

Award marks if answer contains:

- ◆ Two or more substances combine
- ◆ Single new product

Chemical Equation(s):

$CaO + H_2O \rightarrow Ca(OH)_2$ (slaked lime)

EASY **Q7. What is a decomposition reaction? Give one example with a balanced equation.** [1 Mark]

Model Answer:

A reaction in which a single compound breaks down to give two or more simpler substances is a decomposition reaction.

Award marks if answer contains:

- ◆ Single compound breaks down
- ◆ Two or more simpler substances

Chemical Equation(s):

$2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$

EASY **Q8. What is an exothermic reaction? Give one example.** [1 Mark]

Model Answer:
A reaction in which heat is released to the surroundings along with product formation is called an exothermic reaction.

Award marks if answer contains:
- ◆ Heat released
- ◆ Example correct

Chemical Equation(s):

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + Heat$$

EASY **Q9. What is an endothermic reaction? Give one example.** [1 Mark]

Model Answer:
A reaction in which heat is absorbed from the surroundings is called an endothermic reaction.

Award marks if answer contains:
- ◆ Heat absorbed
- ◆ Example correct

Chemical Equation(s):

$$CaCO_3 \rightarrow CaO + CO_2 \text{ (heat absorbed)}$$

EASY **Q10. Define oxidation and reduction in terms of gain or loss of oxygen.** [1 Mark]

Model Answer:
Oxidation: Gain of oxygen OR loss of hydrogen. Reduction: Loss of oxygen OR gain of hydrogen.

Award marks if answer contains:
- ◆ Oxidation = gain of oxygen
- ◆ Reduction = loss of oxygen

EASY **Q11. What is rancidity? How can it be prevented?** [1 Mark]

Model Answer:
Rancidity is the oxidation of fats and oils in food causing unpleasant smell and taste. Prevention: adding antioxidants, flushing with nitrogen, refrigeration, airtight packaging.

Award marks if answer contains:
- ◆ Oxidation of fats/oils
- ◆ Any two prevention methods

EASY **Q12. What is corrosion? Give one example of a metal that undergoes corrosion.** [1 Mark]

Model Answer:

Corrosion is the slow destruction of metals by reaction with air, water or chemicals. Example: Rusting of iron ($Fe_2O_3 \cdot xH_2O$).

Award marks if answer contains:

- ◆ **Slow destruction of metals**
- ◆ **Correct example**

EASY **Q13. Identify the type of reaction: $2Mg + O_2 \rightarrow 2MgO$** [1 Mark]

Model Answer:

Combination reaction and Oxidation reaction (as magnesium gains oxygen).

Award marks if answer contains:

- ◆ **Combination reaction**
- ◆ **Oxidation reaction**

Chemical Equation(s):

$$2Mg + O_2 \rightarrow 2MgO$$

EASY **Q14. Name the reducing agent in the following reaction: $CuO + H_2 \rightarrow Cu + H_2O$** [1 Mark]

Model Answer:

H_2 (hydrogen) is the reducing agent as it gets oxidised (gains oxygen / loses electrons).

Award marks if answer contains:

- ◆ **H2 is the reducing agent**
- ◆ **Reason: H2 gets oxidised**

EASY **Q15. Write the balanced chemical equation for the reaction between iron and steam.** [2 Marks]

Model Answer:

Iron reacts with steam to form iron oxide and hydrogen gas.

Award marks if answer contains:

- ◆ **Correct balanced equation**
- ◆ **Correct products**

Chemical Equation(s):

$$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$$

EASY **Q16. Why is the addition of hydrogen called a reduction reaction? Give one example with a chemical equation.** [2 Marks]

Model Answer:

Addition of hydrogen is reduction because hydrogen is gained by the substance, reducing it. Reduction means gain of hydrogen or loss of oxygen.

Award marks if answer contains:

- ◆ Gain of hydrogen = reduction
- ◆ Correct equation with hydrogen addition

Chemical Equation(s):

$CH_2=CH_2 + H_2 \rightarrow CH_3\text{-}CH_3$ (in presence of Ni catalyst)

EASY **Q17. What is a displacement reaction? Give one example.** [1 Mark]

Model Answer:

A reaction in which a more reactive element displaces a less reactive element from its compound is called a displacement reaction.

Award marks if answer contains:

- ◆ More reactive displaces less reactive
- ◆ Correct example

Chemical Equation(s):

$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

MEDIUM **Q18. Balance the following chemical equation and identify the type of reaction: $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$** [2 Marks]

Model Answer:

Balanced equation and type: Displacement reaction (also redox). Fe_2O_3 is reduced, Al is oxidised. This is the Thermite reaction.

Award marks if answer contains:

- ◆ Correct balanced equation
- ◆ Displacement/redox reaction identified

Chemical Equation(s):

$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$

MEDIUM **Q19. What is a double displacement reaction? How does it differ from a displacement reaction? Give one example of each.** [2 Marks]

Model Answer:

Double displacement: Both reactants exchange ions/radicals. Displacement: Only one element displaces another. 1 mark each for definition + example.

Award marks if answer contains:

- ◆ Correct definition of both
- ◆ One example each with equation

Chemical Equation(s):

Double displacement: $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$

MEDIUM **Q20. Write the balanced chemical equation for the decomposition of lead nitrate on heating. What type of decomposition is this?** [2 Marks]

Model Answer:

Lead nitrate decomposes on heating to give lead oxide, nitrogen dioxide, and oxygen. Type: Thermal decomposition.

Award marks if answer contains:

- ◆ Correct balanced equation
- ◆ Thermal decomposition identified

Chemical Equation(s):

$$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$$

MEDIUM **Q21. What is a precipitation reaction? Explain with the help of an example. Write the balanced chemical equation.** [3 Marks]

Model Answer:

A reaction in which an insoluble precipitate is formed when two solutions are mixed is a precipitation reaction. The precipitate is an insoluble salt.

Award marks if answer contains:

- ◆ Definition correct
- ◆ Insoluble precipitate formed
- ◆ Correct equation

Chemical Equation(s):

$$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$$

MEDIUM **Q22. Explain thermite reaction with a balanced chemical equation. What type of reaction is it and why?** [3 Marks]

Model Answer:

The thermite reaction is the reaction of iron oxide with aluminium. It is a displacement reaction and also highly exothermic, releasing enough heat to melt iron (used in welding).

Award marks if answer contains:

- ◆ Thermite reaction described
- ◆ Displacement + Exothermic identified
- ◆ Correct equation

Chemical Equation(s):

$$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe + Heat$$

MEDIUM **Q23. The colour of copper sulphate solution changes when iron nails are dipped into it. Explain with a balanced chemical equation.** [2 Marks]

Model Answer:

Iron is more reactive than copper and displaces Cu from CuSO4. The blue colour of copper sulphate fades and reddish-brown copper is deposited on the iron nail.

Award marks if answer contains:

- ◆ Iron displaces copper
- ◆ Blue colour fades
- ◆ Correct equation

Chemical Equation(s):

$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

MEDIUM **Q24. Identify the oxidising agent and reducing agent in the following reactions: (a) $CuO + H_2 \rightarrow Cu + H_2O$ (b) $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$** [3 Marks]

Model Answer:

(a) CuO is oxidising agent (gives O to H2); H2 is reducing agent (takes O from CuO). (b) MnO2 is oxidising agent; HCl is reducing agent.

Award marks if answer contains:

- ◆ Correct OA and RA for (a)
- ◆ Correct OA and RA for (b)

MEDIUM **Q25. Why do we apply paint on iron articles? What kind of chemical change are we trying to prevent?** [2 Marks]

Model Answer:

Paint forms a protective barrier preventing iron from reacting with moisture and oxygen in air, thereby preventing corrosion (rusting). The chemical change prevented is: $Fe + O_2 + H_2O \rightarrow Fe_2O_3 \cdot xH_2O$ (rust)

Award marks if answer contains:

- ◆ Barrier prevents O2/H2O contact
- ◆ Corrosion/rusting prevented

MEDIUM **Q26. What happens when silver chloride is kept in sunlight? Write the balanced equation and name the type of reaction.** [3 Marks]

Model Answer:

Silver chloride decomposes in sunlight to form silver (grey) and chlorine gas. This is photolytic decomposition.

Award marks if answer contains:

- ◆ Grey/black silver formed
- ◆ Photolytic decomposition
- ◆ Correct equation

Chemical Equation(s):

$2AgCl \rightarrow 2Ag + Cl_2$ (in presence of sunlight)

MEDIUM **Q27. Why are chips manufacturers usually flush bags of chips with gas like nitrogen? Explain in terms of oxidation.** [2 Marks]

Model Answer:

Nitrogen is an inert gas. Flushing with N2 prevents chips from coming in contact with oxygen, which would oxidise the fats/oils in chips causing rancidity.

Award marks if answer contains:

- ◆ N2 is inert
- ◆ Prevents oxidation
- ◆ Rancidity prevented

MEDIUM **Q28. Write balanced chemical equations for the following: (a) Burning of natural gas (methane) in air (b) Electrolysis of water (c) Reaction of zinc with dilute sulphuric acid** [3 Marks]

Model Answer:

Three separate balanced equations.

Award marks if answer contains:

- ◆ (a) Combustion balanced
- ◆ (b) Electrolysis balanced
- ◆ (c) Zn + H2SO4 balanced

Chemical Equation(s):

(a) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ | (b) $2H_2O \rightarrow 2H_2 + O_2$ | (c) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

MEDIUM **Q29. Define the law of conservation of mass. Verify it using the reaction between sodium sulphate and barium chloride solutions.** [2 Marks]

Model Answer:

Law of Conservation of Mass: Mass can neither be created nor destroyed in a chemical reaction. Verification: Na2SO4 + BaCl2 → BaSO4 + 2NaCl. Total mass of reactants = Total mass of products.

Award marks if answer contains:

- ◆ Correct law statement
- ◆ Verification with equation

Chemical Equation(s):

$Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$

MEDIUM **Q30. What is observed when a few drops of dilute HCl are added to silver nitrate solution? Write the balanced equation.** [2 Marks]

Model Answer:

A white curdy precipitate of AgCl is formed which is insoluble in HNO3. This is a double displacement (precipitation) reaction.

Award marks if answer contains:

- ◆ **White precipitate of AgCl**
- ◆ **Double displacement/precipitation**
- ◆ **Correct equation**

Chemical Equation(s):

$AgNO_3 + HCl \rightarrow AgCl\downarrow + HNO_3$

MEDIUM **Q31. Explain with examples: (a) Thermal decomposition (b) Electrolytic decomposition (c) Photolytic decomposition** [3 Marks]

Model Answer:

(a) Thermal: Heat causes decomposition. (b) Electrolytic: Electric current causes decomposition. (c) Photolytic: Light causes decomposition. 1 mark each with correct example.

Award marks if answer contains:

- ◆ **Thermal decomposition with example**
- ◆ **Electrolytic decomposition with example**
- ◆ **Photolytic decomposition with example**

Chemical Equation(s):

Thermal: $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ | Photolytic: $2AgCl \rightarrow 2Ag + Cl_2$ | Electrolytic: $2H_2O \rightarrow 2H_2 + O_2$

HARD **Q32. How do you distinguish between a physical change and a chemical change? Give two examples of each. Identify which of the following is a chemical change: melting of wax, souring of milk, dissolving salt in water, rusting of iron.** [3 Marks]

Model Answer:

Physical change: No new substance formed, reversible. Chemical change: New substance formed, usually irreversible. Physical: melting of wax, dissolving salt. Chemical: souring of milk, rusting of iron.

Award marks if answer contains:

- ◆ **Correct definitions**
- ◆ **2 examples each**
- ◆ **Correct classification of given 4 changes**

HARD

Q33. What is redox reaction? Explain oxidation and reduction in terms of electron transfer with suitable examples. Identify the oxidising and reducing agents in the reaction: ZnO + C → Zn + CO

[5 Marks]

Model Answer:

Redox: Simultaneous oxidation and reduction. Oxidation: loss of electrons. Reduction: gain of electrons. In ZnO + C → Zn + CO: C is oxidised (reducing agent), ZnO is reduced (oxidising agent).

Award marks if answer contains:

- ◆ Redox definition
- ◆ Electron transfer explanation
- ◆ OA and RA correctly identified

Chemical Equation(s):

$$ZnO + C \rightarrow Zn + CO$$

HARD

Q34. Write the steps you would use to balance a chemical equation. Balance the following equation step by step: $Fe + H_2O \rightarrow Fe_3O_4 + H_2$ Also name the type of reaction and identify oxidising/reducing agents.

[5 Marks]

Model Answer:

Steps: (1) Write word equation. (2) Write skeletal equation. (3) Balance by hit-and-trial. (4) Verify. Balanced: 3Fe + 4H2O → Fe3O4 + 4H2. Type: Redox. Fe is reducing agent, H2O is oxidising agent.

Award marks if answer contains:

- ◆ 4 steps of balancing
- ◆ Correct balanced equation
- ◆ Type identified
- ◆ OA/RA identified

Chemical Equation(s):

$$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$$

HARD

Q35. Why should a magnesium ribbon be cleaned before burning in air? Write the balanced chemical equation for the combustion of magnesium in air. What type of oxides does magnesium form?

[3 Marks]

Model Answer:

Magnesium ribbon is cleaned with sandpaper to remove the layer of magnesium oxide (MgO) from its surface, enabling it to burn properly. Magnesium forms a basic oxide (MgO).

Award marks if answer contains:

- ◆ MgO layer removed by cleaning
- ◆ Correct balanced equation
- ◆ Basic oxide formed

Chemical Equation(s):

$$2Mg + O_2 \rightarrow 2MgO$$

HARD **Q36. Describe any three methods of preventing corrosion of metals. Explain with chemical equations where applicable. Why is iron more prone to corrosion than gold?** **[5 Marks]**

Model Answer:

Methods to prevent corrosion: (1) Painting/oiling – forms a barrier. (2) Galvanisation – coating with zinc. (3) Alloying – making stainless steel. Iron is prone because it readily reacts with O2 and moisture; gold is unreactive (low reactivity in activity series).

Award marks if answer contains:

- ◆ 3 correct methods explained
- ◆ Iron vs gold reactivity compared

Chemical Equation(s):

Rusting: $4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3{\cdot}xH_2O$

HARD **Q37. State the activity series of metals. Using it, predict whether the following reactions will occur or not. Give reasons: (a) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (b) $Cu + FeSO_4 \rightarrow CuSO_4 + Fe$** **[3 Marks]**

Model Answer:

(a) Fe is above Cu in activity series → Fe will displace Cu → Reaction will occur. (b) Cu is below Fe → Cu cannot displace Fe → Reaction will NOT occur.

Award marks if answer contains:

- ◆ Activity series concept applied
- ◆ Correct prediction for (a)
- ◆ Correct prediction for (b) with reason

Chemical Equation(s):

(a) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ (occurs) | (b) $Cu + FeSO_4 \rightarrow$ No reaction

HARD **Q38. Explain the following observations with balanced chemical equations: (a) A white precipitate is formed when sodium sulphate solution is mixed with barium chloride solution. (b) Lime water turns milky when CO_2 is passed through it, but clears up on excess CO_2. (c) A silver article turns black when exposed to air for long.** **[5 Marks]**

Model Answer:

(a) BaSO4 white precipitate – double displacement/precipitation. (b) CO2 + Ca(OH)2 → CaCO3 (milky) → CaCO3 + CO2 + H2O → Ca(HCO3)2 (clear). (c) Ag tarnishes due to Ag2S formation (corrosion).

Award marks if answer contains:

- ◆ (a) BaSO4 precipitate + equation
- ◆ (b) Both equations for milky and clearing
- ◆ (c) Tarnishing = Ag2S

Chemical Equation(s):

(a) $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4\downarrow + 2NaCl$ | (b) $Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$; $CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$ | (c) $4Ag + 2H_2S + O_2 \rightarrow 2Ag_2S + 2H_2O$

HARD

Q39. What is the role of a catalyst in a chemical reaction? Explain the decomposition of hydrogen peroxide in the presence of manganese dioxide with a balanced equation. How does a catalyst affect the rate of reaction? [3 Marks]

Model Answer:

A catalyst increases the rate of a chemical reaction without being consumed. MnO2 acts as catalyst in the decomposition of H2O2: it lowers the activation energy, speeding up the reaction.

Award marks if answer contains:

- ◆ Catalyst definition
- ◆ Correct equation
- ◆ Explains rate increase / activation energy

Chemical Equation(s):

$2H_2O_2 \rightarrow 2H_2O + O_2$ (MnO_2 as catalyst)

HARD

Q40. Distinguish between combination, decomposition, and displacement reactions. Give one balanced equation for each. Also identify which of these reactions are exothermic and which are endothermic, with justification. [5 Marks]

Model Answer:

Combination: A+B→AB (exothermic). Decomposition: AB→A+B (endothermic). Displacement: A+BC→AC+B. Clear distinctions with 1 equation each and correct exo/endothermic classification.

Award marks if answer contains:

- ◆ Combination definition + equation
- ◆ Decomposition definition + equation
- ◆ Displacement definition + equation
- ◆ Exo/endothermic correctly assigned

Chemical Equation(s):

Combination: $CaO + H_2O \rightarrow Ca(OH)_2$ | Decomposition: $CaCO_3 \rightarrow CaO + CO_2$ | Displacement: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

HARD

Q41. A student mistakenly adds excess CO_2 to lime water and notices the milky precipitate dissolves. Write balanced equations for both steps and explain what type of reactions are occurring. [3 Marks]

Model Answer:

Step 1: Ca(OH)2 + CO2 → CaCO3↓ + H2O (milky precipitate – combination/double displacement). Step 2: CaCO3 + H2O + CO2 → Ca(HCO3)2 (clear solution – combination). Both are combination reactions.

Award marks if answer contains:

- ◆ Both balanced equations correct
- ◆ Milky = CaCO3
- ◆ Clear = Ca(HCO3)2
- ◆ Type of reaction for both

Chemical Equation(s):

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$ | $CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$

HARD

Q42. Analyse the following statement: 'All combustion reactions are combination reactions but all combination reactions are not combustion reactions.' Justify with two examples each and relevant balanced equations.

[5 Marks]

Model Answer:

Combustion is always combination (fuel combines with O2). But combination need not involve combustion (e.g., CaO + H2O, no burning). Full marks for 2 combustion + 2 non-combustion combination examples with equations and justification.

Award marks if answer contains:

- ◆ All combustion = combination (justified)
- ◆ Not all combination = combustion (justified)
- ◆ 2 combustion examples
- ◆ 2 non-combustion combination examples

Chemical Equation(s):

Combustion: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$; $2Mg + O_2 \rightarrow 2MgO$ | Non-combustion combination: $CaO + H_2O \rightarrow Ca(OH)_2$; $SO_3 + H_2O \rightarrow H_2SO_4$